⍰⍰ SUVIDHA ONE

Additional Technical Documentation

Final Justification & Solution Design

SUBMISSION DETAILS	
Team Name	The_Dark_Knight
Team ID	21
Project	SUVIDHA ONE – Unified Citizen Service Kiosk
Stage	Prototype Ready ✓
Date	February 2026
GitHub Repository	github.com/jadavmadhavkumar/SUVIDHA_ONE
Drive / Demo Assets	Google Drive Folder

"One Kiosk, All Services – Suvidha Sabke Liye"

SECTION 1 — Project Maturity & Development Status

1. Project Maturity & Development Status

1.1 Current Development Stage

✓/☐	STAGE	DESCRIPTION
☐	Idea Only	*Initial concept phase — not applicable*
✓	**Prototype Ready**	*Core kiosk UI screens built; authentication, bill payment and complaint flows functional in demo environment*
☐	MVP Developed	*Full MVP with all integrations — in progress (Sprint 3)*
☐	Fully Functional	*Production deployment with live government API integrations — Phase 2 target*

1.2 What is Built & Working (Prototype)

COMPLETED FEATURES	IN PROGRESS / NEXT SPRINT
• Welcome screen with animated branding • Language selection (Hindi + English UI) • OTP-based authentication flow (mock SMS) • Service dashboard with icon grid • Electricity bill fetch & payment UI (BBPS sandbox) • Gas cylinder booking flow • Municipal complaint filing form • Receipt generation (PDF + print preview) • Voice guidance (TTS — Hindi & English) • High-contrast & large-font accessibility modes	• Water board bill integration (live API) • Aadhaar biometric authentication (UIDAI sandbox) • DigiLocker certificate module • UPI QR payment (NPCI sandbox) • Tamil & Bengali language packs • Offline sync engine • Admin monitoring dashboard • WhatsApp receipt delivery

1.3 Project Links

RESOURCE	LINK / DETAILS
GitHub Repository	https://github.com/jadavmadhavkumar/SUVIDHA_ONE
Demo Assets / Drive	https://drive.google.com/drive/folders/1GdXYJ1P3-cR9KIBXeaCLq_wmfmUjq168
Video Demo	Available in Google Drive folder above
Tech Stack (Frontend)	React 18 + Next.js 14, TypeScript, TailwindCSS
Tech Stack (Backend)	Node.js 20 LTS, PostgreSQL 16, Redis 7, Docker
Kiosk OS	Ubuntu 22.04 LTS — Kiosk Mode (locked browser)

SECTION 2 — Department-wise Features in Kiosk Touch Interface

2. Department-wise Features in Kiosk Touch Interface

SUVIDHA ONE integrates 6 service departments under a unified touch interface. Each department's features are specifically designed for self-service kiosk usage — pictorial navigation, voice guidance, minimal text input, and one-tap actions wherever possible.

⚡ ELECTRICITY

FEATURE	DESCRIPTION
View & Pay Bill	Fetch current outstanding bill via BBPS; display amount, billing period, due date; single-tap payment via UPI/Card/Cash
New Connection Apply	Fill application form; upload ID via kiosk camera; get reference number; track status by application ID
Load Enhancement	Request for increased sanctioned load; auto-calculates charges; submit supporting documents digitally
Outage / Complaint	Report power cut, meter fault, wire breakage with location pin; receive complaint token; track resolution timeline
Meter Reading Submission	Self-service meter reading upload via camera; system validates against historical range before submission
Usage History & Alerts	View 12-month consumption graph; set usage alert thresholds; compare with neighbourhood average
Name Transfer / Change	Apply for ownership transfer post property sale; upload required legal documents; status tracking
Duplicate Bill Print	Print/download duplicate bill for any past 24 months; QR-verified official copy

⛽ GAS / LPG

FEATURE	DESCRIPTION

Cylinder Booking	Book refill cylinder with one tap; select preferred delivery slot; SMS/WhatsApp delivery confirmation sent to registered mobile
Booking Status Check	Track active booking status (Booked → Dispatched → Delivered) in real time via distributor API
Emergency Gas Leak Report	High-priority complaint with auto-alert to distributor + GAIL emergency helpline; location tagged
KYC Update / eKYC	Update mobile number, address, bank account for subsidy via Aadhaar-based eKYC — no office visit required
Subsidy Status Check	View PAHAL subsidy credit history; check if DBT transfer is active; resolve mismatch complaints
New Connection Apply	Apply for new LPG connection; check eligibility; upload Aadhaar + address proof; get connection ID
Surrender Connection	Digitally surrender duplicate/unused connection; receive confirmation receipt for records
Distributor Locator	Find nearest LPG distributor with address, timings, contact using Google Maps integration

WATER SUPPLY

FEATURE	DESCRIPTION
View & Pay Water Bill	Fetch water board bill by consumer number or Aadhaar; pay via BBPS-integrated gateway; instant receipt
Leakage / Pipe Burst Report	Report leakage or burst main with photo evidence via kiosk camera; auto-priority based on road/area type
New Water Connection	Apply for domestic/commercial connection; upload property documents; track application stage
Sewage / Drainage Complaint	Report blocked drain or sewage overflow; attach geo-tagged photo; escalation if unresolved in 48 hours
Water Quality Complaint	Report discoloured or contaminated supply; trigger automatic sampling request from waterboard lab
Meter Replacement Request	Apply for faulty meter replacement; upload meter photo as evidence; estimated timeline displayed

Tanker Water Request	Emergency water tanker booking for areas under supply disruption; confirm availability & slot
Consumption History	View monthly consumption; detect sudden spike; get conservation tips in local language

▯ MUNICIPAL CORPORATION

FEATURE	DESCRIPTION
Property Tax Payment	Fetch property tax demand via property ID/owner name; view calculation breakup; pay full or installment; print challan
Trade License Renewal	Renew annual trade/shop license; upload updated documents; pay fee; get digitally signed license
Birth Certificate	Apply for new birth certificate or reprint; integrated with hospital birth records; print on spot or download
Death Certificate	Register or obtain death certificate; hospital/home integration; required for legal/financial proceedings
Marriage Certificate	Apply for new or certified copy of marriage certificate; track application; download/print when ready
Building Plan Approval Status	Check approval status of submitted building plans by application number; get officer remarks
Solid Waste / Garbage Complaint	Report missed garbage collection, illegal dumping, overflowing bin; photo upload; ward-wise tracking
Street Light Complaint	Report defunct street lights by location; auto-assigns to nearest ward maintenance team
Public Toilet / Infrastructure	Report damaged public amenities; geo-tagged complaints forwarded to ward officer's dashboard
Encroachment Complaint	Anonymously or by name report road/footpath encroachment; confidential case tracking

▯ GOVERNMENT CERTIFICATES & DOCUMENTS

FEATURE	DESCRIPTION

Income Certificate	Apply online with Aadhaar verification; upload bank/salary proof; track approval; print signed copy
Caste / Community Certificate	Apply for SC/ST/OBC certificate; Aadhaar-based identity; officer digital signature; DigiLocker save
Domicile / Residence Certificate	Verify residence via Aadhaar address; get digitally signed domicile certificate; instant print
DigiLocker Integration	Access existing documents (Driving License, Aadhaar, Marksheet, PAN) stored in DigiLocker; print official copies
Document Verification	Verify authenticity of any government document by scanning its printed QR code at kiosk

⍰ GRIEVANCE & COMPLAINT MANAGEMENT

FEATURE	DESCRIPTION
Cross-Department Complaint Filing	Single window for filing complaint against any department — Electricity, Water, Gas, Municipal
Real-Time Status Tracking	Track complaint via unique ID; view timeline (Filed → Assigned → In Progress → Resolved)
Escalation Engine	Auto-escalates complaint to senior officer if not resolved within SLA (48h standard / 24h emergency)
Satisfaction Rating	Post-resolution 1–5 star rating + optional voice/text feedback; contributes to officer performance metrics
Complaint History	View all past complaints and resolutions in current session; SMS history accessible any time

SECTION 3 — UI/UX Suitability for Touch-Based Kiosk Interface

3. UI/UX Suitability for Touch-Based Kiosk Interface

YES — The SUVIDHA ONE interface is purpose-built exclusively for touch-based kiosk usage.

Every design decision — from button sizes and icon placement to font sizes and screen flow — has been made with the constraints of a public-facing, large-screen touch kiosk in mind, not a desktop browser or mobile app.

3.1 Touch Interaction Design Constraints & Solutions

CONSTRAINT	DESIGN CHALLENGE	OUR SOLUTION
Fat-Finger / Mis-Tap	Large fingertips easily mis-tap small targets on 32–55" screens	Minimum button size: 60×60mm (≥160px at 72dpi). 12–16mm gap between adjacent targets. Rounded corners signal tappability.
No Hover State	Desktop UI relies on hover to show interactivity — unavailable on touch	All interactive elements have distinct pressed/active state (colour darken + scale). No hover-only tooltips.
Glare & Visibility	Outdoor/indoor kiosk screens often have glare from lighting or sunlight	1000+ nit anti-glare display. High contrast color scheme (WCAG AA ≥ 4.5:1). Dark icons on light backgrounds.
Screen Distance	Users stand 40–70cm from screen (arm's length), not 30cm like a phone	Minimum font size 48px body / 72px headings. Icons minimum 120×120px. Service labels max 2 words.
No Physical Keyboard	Users unfamiliar with virtual keyboards; typing is slow and error-prone	Minimize text input. Use numeric pads for mobile/OTP. Aadhaar/QR auth eliminates typing. Voice input as alternative.
Unintended Screen Press	Users may lean on screen or touch accidentally while reading	300ms debounce on all taps. Confirmation modal for all financial

		actions. Undo within 5 seconds for non-payment.
Dirty / Wet Hands	Public kiosk users may have dusty, wet, or differently-textured hands	Capacitive touch sensitivity tuned at hardware level. Large targets reduce need for precision. Stylus compatible.
Multi-User Sessions	Each citizen needs a clean slate; shared device must not leak session data	Hard 3-min inactivity auto-logout. Session clear on logout. No data cached in browser localStorage.
Screen Height / Reach	Tall service tiles at top of screen unreachable for short users or wheelchair users	Bottom-centric layout — primary CTAs always in lower 60% of screen. Height-adjustable stand (700–1200mm).
Limited Options Per Screen	Too many choices paralyze users, especially low-literacy or elderly	Max 6 service tiles per grid. Wizard-style step-by-step flows. Never more than 4 options in a decision.

3.2 Kiosk-Specific UI Patterns

- ICON-DOMINANT LAYOUT: Service tiles show a large icon (120–180px) above a maximum 2-word label — usable without reading
- WIZARD FLOW: Every service follows a linear step-by-step wizard (1 action per screen) — no scrolling, no multi-field forms
- PERSISTENT NAVIGATION: Home | Back | Help | Logout buttons always visible in bottom nav bar — users never get lost
- VOICE-SYNCED TEXT: On-screen text is simultaneously read aloud in selected language — dual-channel communication
- LARGE NUMERIC KEYPAD: Custom on-screen keypad (keys 80×80px minimum) for OTP/mobile number entry
- COLOUR-CODED STATUS: Green = success/payment, Red = error/complaint, Orange = pending/warning — universal recognition
- CONFIRMATION SCREENS: Every payment/submission shows full summary before final action — prevents costly mistakes
- ANIMATED FEEDBACK: Tap animations, loading skeletons, and progress bars reassure users that the system is responding

SECTION 4 — Deployment & Practical Feasibility

4. Deployment & Practical Feasibility

4.1 Deployment Summary

PARAMETER	SPECIFICATION
Target Deployment Model	HYBRID — On-Premise kiosk node + Cloud backend services
Cloud Platform	AWS India (Mumbai ap-south-1) + Azure Government Cloud (fallback)
On-Premise Component	Kiosk device runs React/Next.js app locally; transactions synced to cloud
Internet Dependency	MEDIUM — High-speed not required; 2 Mbps per kiosk sufficient
Offline Mode Support	YES — Full offline capability for status checks, bill viewing, form pre-fill
Connectivity Priority	4G LTE Primary → Wi-Fi Secondary → Ethernet Fallback → Offline Mode
Deployment Locations	Gram Panchayat, Urban Local Body, Govt Office, Railway Station, Hospital, CSC
Rollout Model	Pilot (5 kiosks) → City (50 kiosks) → State (500+) → National fleet
Setup Time per Kiosk	< 4 hours — plug-and-play with pre-configured disk image
Remote Management	Centralized admin dashboard; OTA updates; remote reboot; health monitoring

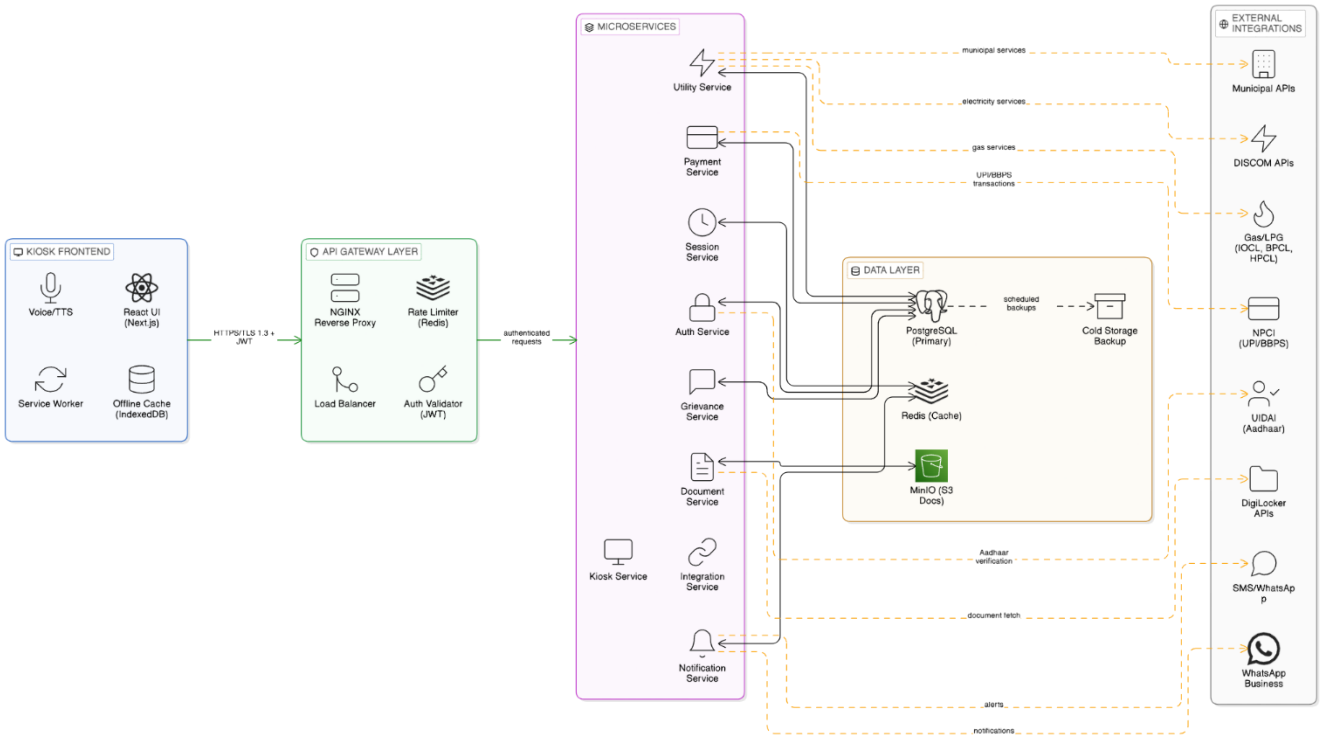
4.2 Infrastructure Requirements

COMPONENT	SPECIFICATION	COST ESTIMATE (CAPEX)	NOTES

Kiosk Hardware	32" PCAP touch, i5 mini-PC, 8GB RAM, 256GB SSD, thermal printer	**₹80,000–1,20,000 per unit**	*IP54 enclosure; 5-year lifespan*
Connectivity	4G SIM + Ethernet port; monthly data ~5 GB/kiosk	**₹500–800/month OPEX**	*Jio/BSNL government tariff*
Cloud Backend (shared)	2× AWS t3.medium (API), 1× RDS db.t3.small (DB), ElastiCache	**₹8,000–12,000/month for 50 kiosks**	*Scales linearly; shared across fleet*
CDN & Storage	CloudFront + S3 (icons, fonts, receipts)	**₹1,000–2,000/month**	*Reduces kiosk bandwidth significantly*
Solar Backup (optional)	200W solar panel + 100Ah battery for off-grid sites	**₹15,000–25,000 one-time**	*Phase 3 addition for remote kiosks*
Maintenance SLA	Annual maintenance contract with hardware vendor	**₹8,000–12,000/year per unit**	*Covers parts, on-site support*

4.3 Offline Mode — Detailed Capability

FEATURE	OFFLINE BEHAVIOUR	SYNC BEHAVIOUR
Bill Viewing	Last fetched bill shown from IndexedDB cache (72-hour TTL)	Refreshes automatically on reconnect
Complaint Filing	Form data saved locally; submitted to queue	Queue flushes to server within 30 seconds of reconnect
Authentication	OTP not possible offline; QR / cached token used if valid session	New logins resume normally on reconnect
Service Info / FAQs	Full service catalogue cached as Service Worker asset	Updated on app version update
Payments	BLOCKED in offline mode — financial transactions require real-time API	User informed clearly; receipt issued post-reconnect
Certificate Status	Last known status shown with 'Last Updated' timestamp visible	Live status fetched on reconnect
Offline Indicator	Persistent orange 'Offline Mode' banner shown on all screens	Banner disappears when connectivity restored


KIOSK FRONTEND
Voice/TTS
React UI (Next.js)
Service Worker
Offline Cache (IndexedDB)
HTTPS/TLS 1.3 + JWT
API GATEWAY LAYER
NGINX Reverse Proxy
Rate Limiter (Redis)
Load Balancer
Auth Validator (JWT)
authenticated requests
MICROSERVICES
Utility Service
Payment Service
Session Service
Auth Service
Grievance Service
Document Service
Kiosk Service
Integration Service
Notification Service
DATA LAYER
PostgreSQL (Primary)
scheduled backups
Cold Storage Backup
Redis (Cache)
MinIO (S3 Docs)
municipal services
electricity services
gas services
UPI/BBPS transactions
Aadhaar verification
document fetch
alerts
notifications
EXTERNAL INTEGRATIONS
Municipal APIs
DISCOM APIs
Gas/LPG (IOCL, BPCL, HPCL)
NPCI (UPI/BBPS)
UIDAI (Aadhaar)
DigiLocker APIs
SMS/WhatsApp
WhatsApp Business

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SECTION 5 — Accessibility & Inclusion Details

5. Accessibility & Inclusion Details

SUVIDHA ONE is designed with 'Inclusion by Default' — every citizen, regardless of disability, literacy level, age, language, or digital experience, must be able to use the kiosk independently.

5.1 Accessibility Support Matrix

USER GROUP	SUPPORT	IMPLEMENTATION DETAIL
Visually Impaired	**✓Full Support**	TTS reads all on-screen text in selected language; all icons have ARIA labels; screen reader (NVDA/JAWS) compatible audio output; high-contrast mode (white-on-black + yellow accents); font size up to 150% of base
Hearing Impaired	**✓Full Support**	All audio guidance has on-screen text equivalent; visual confirmation animations replace audio-only cues; no audio-only error states; captions on any video content
Senior Citizens (60+)	**✓Full Support**	Senior Mode: simplified 3-tile home screen (Pay Bill / File Complaint / Get Certificate); extra-large fonts (150%); slow-paced voice guidance; longer timeout (6 minutes); confetti/success animation for reassurance
Low-Literacy Users	**✓Full Support**	Icon-first design usable without reading; pictorial service flows; voice reads every option aloud before action; Hindi numerals option; no jargon — plain language throughout
Wheelchair Users	**✓Full Support**	Height-adjustable kiosk stand (700–1200mm); all primary CTAs in bottom 60% of screen; no content requires reaching above 1200mm; seated-use tested in prototype
Non-Smartphone Users	**✓Full Support**	No personal device required; OTP delivered via SMS to basic phone; physical receipt always available; QR code on receipt scannable later for status check
Rural / First-time Users	**✓Full Support**	Guided tutorial on first use (skippable); help agent call button always visible; voice-explained each step

		before citizen must act; 'How to use this kiosk' demo loop on idle screen
Differently-abled (Motor)	**✓Partial**	Large touch targets (60×60mm min); no drag/swipe required — all tap-based; stylus compatible; Phase 2: switch-access support for users with limited motor control

5.2 Regional Language Support

LANGUAGE	SCRIPT	STATUS	VOICE (TTS)
Hindi	Devanagari	✓Live in Prototype	✓Google TTS — Hindi (India)
English	Latin	✓Live in Prototype	✓Google TTS — English (India)
Tamil	Tamil Script	⍰ Sprint 6 (In Progress)	✓Google TTS — Tamil
Bengali	Bengali Script	⍰ Sprint 6 (In Progress)	✓Google TTS — Bengali
Telugu	Telugu Script	⍰ Phase 2 Q3 2026	✓Available via API
Kannada	Kannada Script	⍰ Phase 2 Q3 2026	✓Available via API
Marathi	Devanagari	⍰ Phase 2 Q3 2026	✓Available via API
Gujarati	Gujarati Script	⍰ Phase 2 Q4 2026	✓Available via API
Odia	Odia Script	⍰ Phase 2 Q4 2026	✓Available via API
Punjabi	Gurmukhi	⍰ Phase 3	✓Available via API

Font Implementation: Noto Sans Devanagari for Hindi/Marathi; Noto Sans Tamil/Bengali/Telugu etc. via Google Fonts CDN. All fonts subset to reduce load time. CDAC Gargi/Shruti as government-approved fallbacks.

5.3 UI Compliance Standards

STANDARD	COMPLIANCE LEVEL	NOTES
WCAG 2.1	**Level AA ✓**	Contrast ≥ 4.5:1; keyboard/voice navigation; ARIA labels on all interactive elements; no seizure-inducing animations

RPwD Act 2016	**Compliant ✓**	Rights of Persons with Disabilities Act — physical kiosk height adjustment + digital accessibility features
Digital India UI Guidelines	**Aligned ✓**	National color scheme, government font recommendations (Noto/CDAC), branding guidelines followed
BIS IS 17802	**In Progress ⏳**	Indian Standard for accessible ICT — accessibility checklist being completed pre-submission
GIGW 3.0	**Aligned ✓**	Guidelines for Indian Government Websites — e-governance UI/UX standards applied to kiosk interface
ISO 9241 (Ergonomics)	**Aligned ✓**	Touchscreen interaction ergonomics — target sizes, spacing, feedback latency all within ISO specifications

SECTION 6 — Security Architecture & Design

6. Security Architecture & Design

Security Philosophy: SECURITY BY DESIGN — NOT AN AFTERTHOUGHT

Security is embedded at every layer of SUVIDHA ONE from day one of architecture design. We follow a Zero-Trust model — no component trusts any other by default; every request is authenticated, authorized, encrypted, and logged. Citizen PII is never stored on kiosk hardware; all sensitive data flows through encrypted channels to government-grade secured backends.

6.1 Overall Security Architecture

SECURITY LAYER	CONTROLS IMPLEMENTED	STANDARD / FRAMEWORK
L1 — Physical Kiosk	Secure boot (UEFI + TPM 2.0); full-disk encryption (LUKS/BitLocker); tamper-evident seals; CCTV integration; USB ports disabled via BIOS; kiosk OS lockdown (no external access)	*FIPS 140-2 (disk encryption); BIS hardware security standards*
L2 — Transport	TLS 1.3 mandatory for all API calls; certificate pinning on kiosk app (prevents MITM); HSTS headers; no HTTP fallback; API calls via private VPC endpoints where possible	*OWASP Top 10 A02; NIST SP 800-52*
L3 — Identity & Auth	JWT tokens (15-min expiry) + refresh rotation; Aadhaar OTP via certified AUA channel; no Aadhaar number stored anywhere; session token invalidated immediately on logout	*UIDAI AUA/KUA Circular; OAuth 2.0 RFC 6749; IS 17428*
L4 — Data Encryption	AES-256-GCM at rest (PostgreSQL pgcrypto extension); field-level encryption for mobile number, transaction ID; tokenization for payment card data via PCI-DSS compliant gateway	*DPDP Act 2023; IT Act 43A; PCI-DSS v4.0*
L5 — PII Protection	Aadhaar number never stored — UIDAI tokenization (VID) used; minimal data	*DPDP Act 2023; UIDAI Data Protection Policy*

	collection principle; explicit consent screen before any data collection; right-to-erasure implemented	
L6 — Payment Security	No card/account data processed on kiosk; redirected to PCI-DSS Level 1 certified payment gateway (BBPS/NPCI); 3D Secure for card transactions; UPI VPA resolution via NPCI only	*PCI-DSS v4.0; RBI Payment Aggregator Guidelines 2021*
L7 — API Security	Kong API Gateway: rate limiting (100 req/min/IP), JWT validation, request size limits, IP allowlisting for government APIs; WAF (AWS WAF) rules for OWASP Top 10 attacks	*OWASP API Security Top 10; NIST SP 800-95*
L8 — Application Security	Input validation & sanitization at every field (server-side, not just client); SQL parameterized queries only (no ORM string interpolation); XSS/CSRF protection; Content Security Policy headers	*OWASP Top 10 A03/A07; CWE-79/89*
L9 — Audit & Logging	Immutable audit trail for all transactions, auth events, admin actions — write-once append-only PostgreSQL partition; log forwarding to SIEM (ELK); 2-year retention	*DPDP Act; CERT-In Directions 2022; CAG audit readiness*
L10 — Vulnerability Management	Automated SAST (Snyk) on every git push; DAST (OWASP ZAP) monthly; dependency vulnerability scan (Dependabot); quarterly external penetration testing; CVE patching SLA: Critical ≤24h, High ≤7 days	*CERT-In Guidelines; NCSC Cyber Essentials; ISO 27001*

6.2 Security Integration: By Design vs Afterthought

Every architectural decision in SUVIDHA ONE was made with security as a first-class requirement:

SECURITY BY DESIGN EVIDENCE	WHAT THIS PREVENTS
Aadhaar token (VID) used from Day 1 — Aadhaar number collection never designed in	Eliminates the largest PII breach risk from the start
Zero-Trust architecture — all inter-service calls require auth tokens	Prevents lateral movement if one service is compromised
Field-level encryption designed into DB schema, not added later	Data at rest is protected even if DB dump is exfiltrated

Session data never in localStorage — always server-side only	Prevents XSS-based session hijacking at kiosk
Payment redirected to BBPS/NPCI gateway — no card data touches our server	Eliminates entire PCI-DSS scope from our application
Immutable audit logs from Sprint 1 — not a release-time addition	Evidence trail available from day one; no log gaps
HTTPS-only, certificate pinning in kiosk app from first release	Man-in-the-middle attacks impossible from kiosk hardware
Rate limiting designed into API Gateway, not application code	DDoS/brute-force protection at infrastructure level

6.3 Threat Modelling

YES — Formal threat modelling conducted using the STRIDE framework (Microsoft) and OWASP Threat Dragon tool.

STRIDE THREAT	ATTACK SCENARIO IDENTIFIED	MITIGATION DESIGNED	RESIDUAL RISK
Spoofing	Attacker impersonates citizen using stolen OTP	TOTP + OTP hybrid; device fingerprinting; anomaly detection on auth location	**LOW — multi-factor mitigates**
Tampering	Attacker modifies in-transit API request to alter bill amount	Request signing (HMAC-SHA256) on all financial API calls; server-side amount validation	**LOW — server always canonical**
Repudiation	Citizen denies making payment; merchant disputes transaction	Immutable audit trail; BBPS generates independent transaction ID; SMS receipt to citizen	**VERY LOW — dual-party trail**
Info Disclosure	Aadhaar or mobile number leaked via browser cache or logs	No PII in URLs, logs, or localStorage; field-level DB encryption; cache-control headers	**LOW — encryption mitigates**
Denial of Service	Flood of kiosk requests overwhelms backend API	Kong rate limiting 100 req/min; AWS WAF DDoS protection; Circuit breaker (Resilience4j)	**LOW — auto-scaled backend**
Elevation of Priv	Kiosk user gains admin access via URL manipulation	Role-based access control (RBAC); kiosk runs with no admin token; locked OS; secure boot	**VERY LOW — OS lockdown**

Physical Tampering	Attacker installs keylogger or card skimmer on kiosk	Tamper-evident hardware seals; CCTV with AI anomaly detection; no card reader on kiosk	**LOW — hardware controls**
Supply Chain Attack	Malicious package in npm dependencies	Snyk SCA on every build; package pinning in package-lock.json; private npm registry proxy	**MEDIUM — ongoing monitoring**

6.4 Regulatory Compliance

- DPDP Act 2023: Consent-first data collection; data minimization; breach notification ≤72 hours; right to erasure implemented
- UIDAI AUA/KUA Compliance: Licensed AUA integration; no raw Aadhaar storage; VID-based tokenization; demographic data handling per UIDAI circular
- IT Act 2000 (Sec 43A & 72A): Reasonable security practices; information privacy; cyber incident readiness plan in place
- RBI Payments Guidelines 2021: BBPS-integrated payments; settlement timelines; dispute resolution workflow designed
- CERT-In Directions 2022: Mandatory incident reporting ≤6 hours; log retention ≥180 days; server NTP time synchronization
- PCI-DSS v4.0: Payment fully offloaded to certified PSP; no card data on our infrastructure; scope minimized

SUVIDHA ONE — End of Additional Technical Documentation

Team: The_Dark_Knight | ID: 21 | February 2026

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